Statistics of Financial Markets

Chapter 3: Efficient Market Hypothesis

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Outline

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Introduction

Weak-Form Efficiency

Semi-Strong Efficiency

Strong-Form Efficiency

Empirical Evidence

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Motivation

- Why efficient markets matter for financial decision-making
- Historical development of the Efficient Market Hypothesis (EMH)
- Implications for portfolio management and asset pricing

Introduction

- Formal definition of market efficiency
- Fama's seminal contributions (1965, 1970)
- Information sets and price formation

Weak-Form Efficiency

- Prices reflect all past trading information
- Implications: technical analysis cannot generate excess returns
- Tests: autocorrelation, runs tests, variance ratio tests

Semi-Strong Efficiency

- Prices reflect all publicly available information
- Implications: fundamental analysis cannot generate excess returns
- Tests: event studies, earnings announcements

Strong-Form Efficiency

- Prices reflect all information, including private/insider information
- Implications: no investor can consistently earn excess returns
- Tests: insider trading studies, mutual fund performance

Empirical Evidence

- Evidence for and against EMH across global markets
- Anomalies: momentum, value effect, calendar effects
- Behavioral finance challenges to EMH

SFM_ch3_emh_tests

Conclusions

- Summary of EMH forms and their testable implications
- Current state of the debate
- Practical relevance for investors and regulators

References

- Fama, E.F. (1970). Efficient capital markets: A review of theory and empirical work.
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- Malkiel, B.G. (2003). The efficient market hypothesis and its critics.